2.2 Covariance and correlation matrices. Linear algebra.

1. Show that any covariance matrix is symmetric positive semidefinite. Show that the same is true for correlation matrices.

2. Find the correlation matrix of three random variables with covariance matrix

$$\Sigma_X = \begin{pmatrix} 1 & 0.36 & -1.44 \\ 0.36 & 4 & 0.80 \\ -1.44 & 0.80 & 9 \end{pmatrix}.$$

3. Assume that all the entries of an $n \times n$ correlation matrix which are not on the main diagonal are equal to ρ. Find upper and lower bounds on the possible values of ρ.

4. How many eigenvalues does an $n \times n$ matrix with real entries have? How many eigenvectors?

5. Let

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}.$$

(i) Find a 2×2 matrix M such that $M^2 = A$;

(ii) Find a 2×2 matrix M such that $A = MM^t$.

6. The 2×2 matrix A has eigenvalues 2 and -3 with corresponding eigenvectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$. If $v = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, find Av.

7. Let A and B be square matrices of the same size. Show that the traces of the matrices AB and BA are equal.

8. Can you find $n \times n$ matrices A and B such that

$$AB - BA = I_n,$$

where I_n is the identity matrix of size n?

9. A probability matrix is a matrix with nonnegative entries such that the sum of the entries in each row of the matrix is equal to 1. Show that the product of two probability matrices is a probability matrix.

10. Find all the values of ρ such that

$$\begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix}$$

is a correlation matrix.[1]

[1] A solution to this question was given in Chapter 1 using Sylvester's criterion; two different solutions will be given herein.